COMPSCI 130 S1 2022
Introduction to Software Fundamentals

Binary Search Trees – Introduction

Agenda & Reading

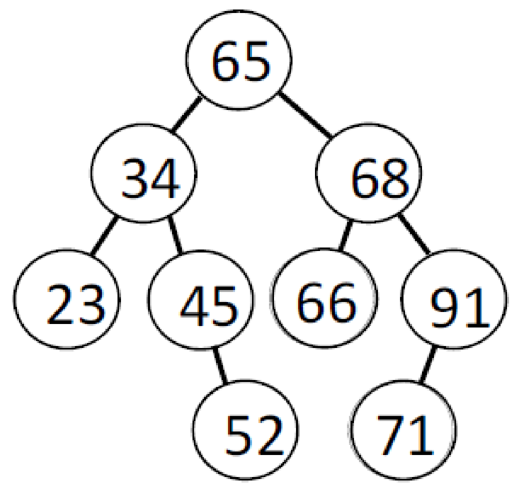
- Agenda
  - What is a binary search tree?
  - Inserting nodes into a binary search tree
- Online Coursebook:
  - https://onlinetextbook.auckland.ac.nz/11-binary-search-trees/

Trees can be efficient

- Trees can be efficient.
  - Efficiency usually dependant on the height of the tree.
  - Tree-based algorithms can work in O(log n) time.

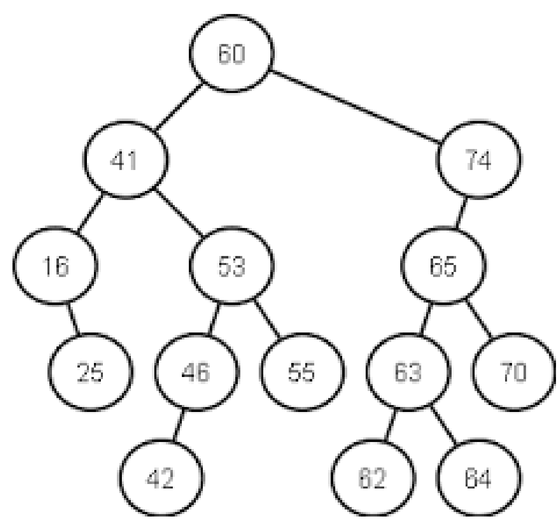
- Can we use trees for searching and sorting?
  - Observation: With a sorted collection of items, we can search for a particular item efficiently using binary search.
  - Idea: Design trees which define order!

23	34	45	52	65	66	68	71	91



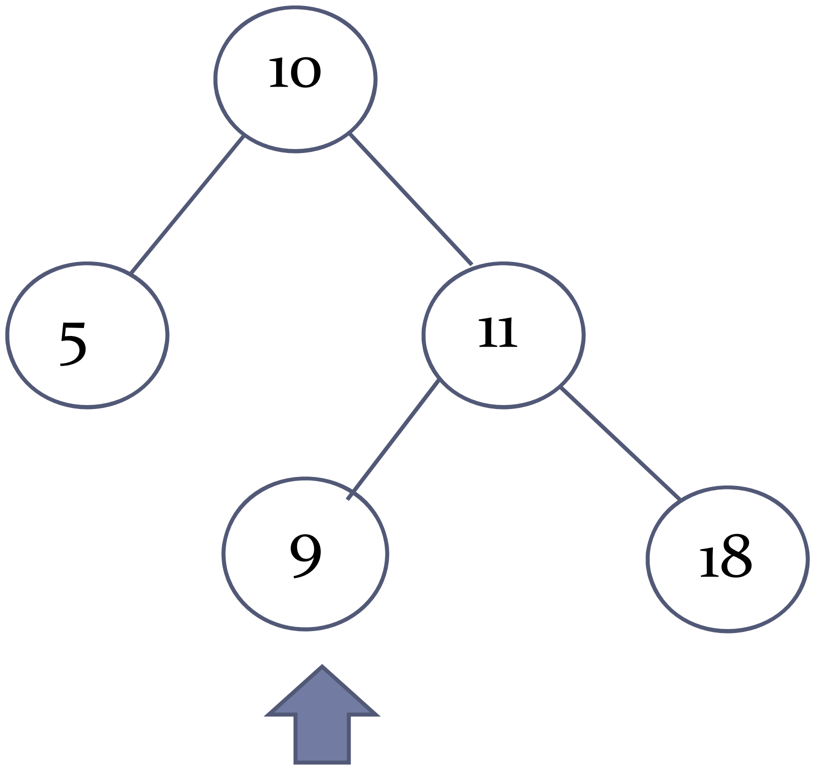
Binary search trees

- Binary search trees are binary trees with the following properties:
  1. For all nodes, the values in the left subtree of that node are smaller than the value of the node.
  2. For all nodes, the values in the right subtree of that node are greater than the value of the node.



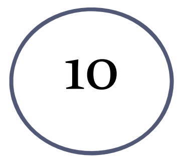
Is the following a binary search tree?

No



Is the following a binary search tree?

Yes



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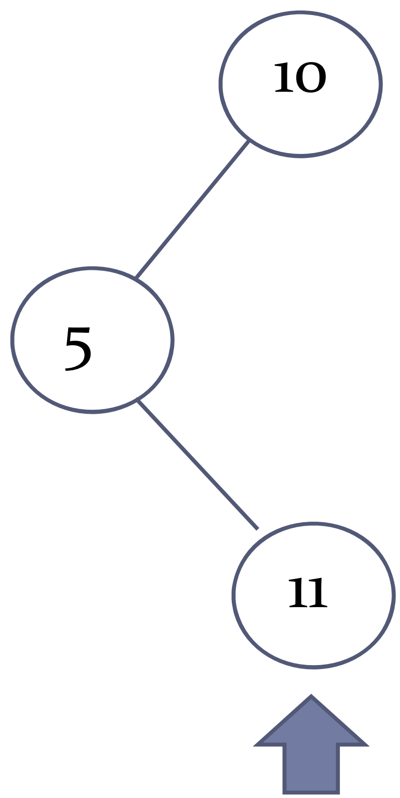

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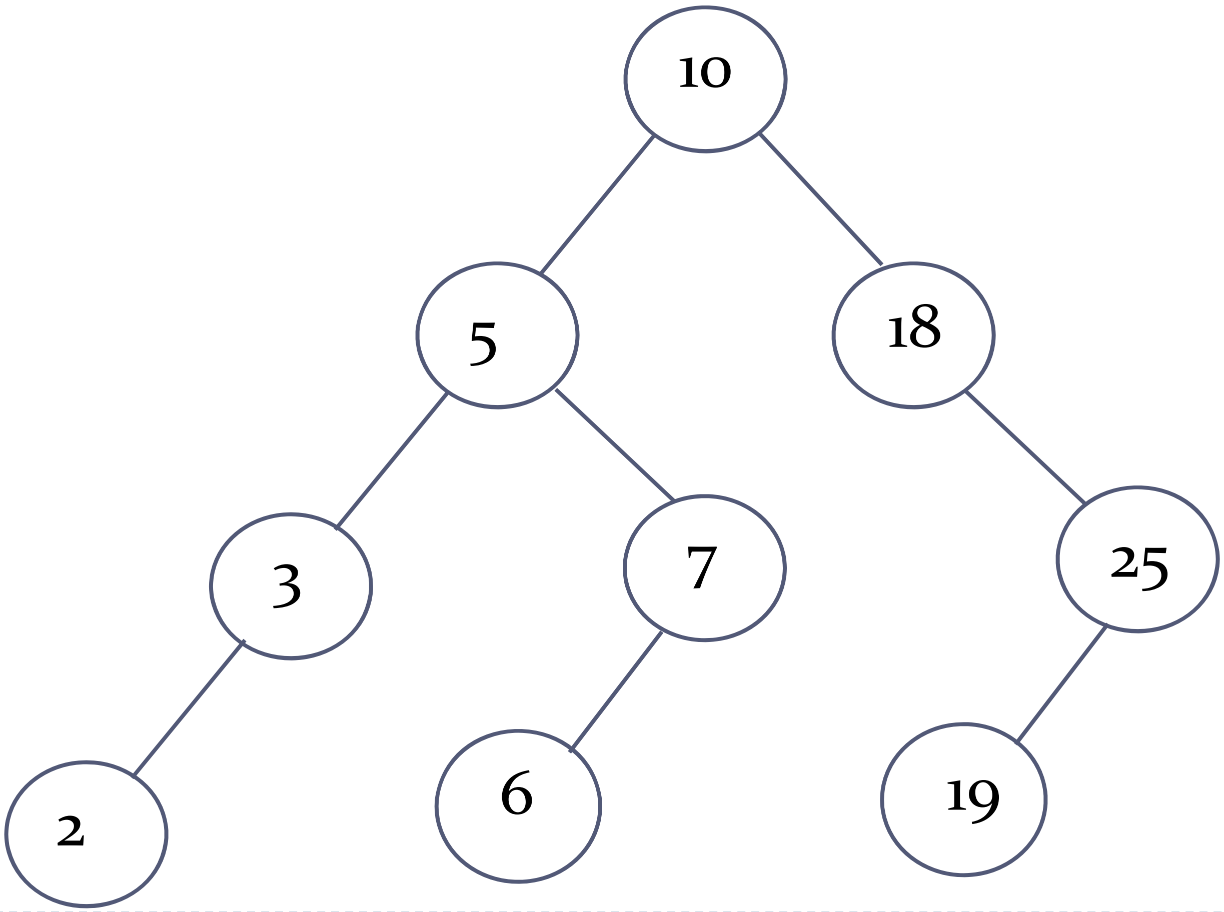
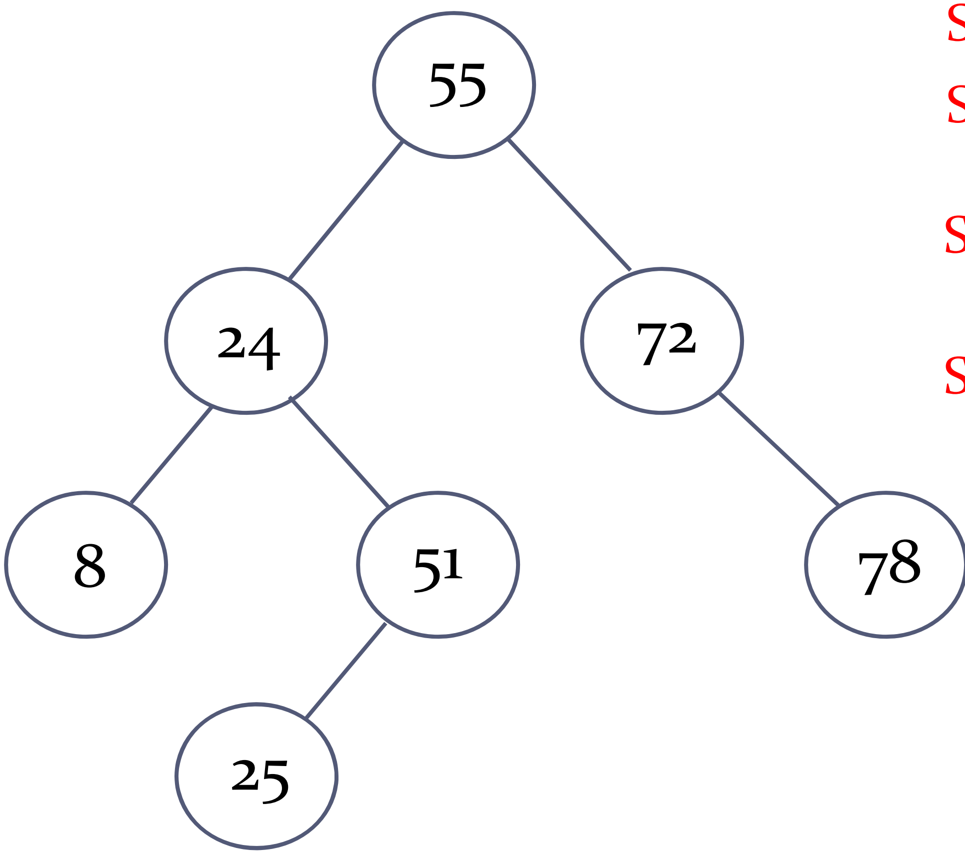
Yes



Inserting nodes into a binary search tree

- How can we insert a node into a binary search tree, such that the binary search tree property is maintained?



Step 1: compare node with root

Step 2: if node < root, go left
if node > root, go right

Step 3: repeat Step 2 until we reach an empty spot

Step 4: insert the node at that spot

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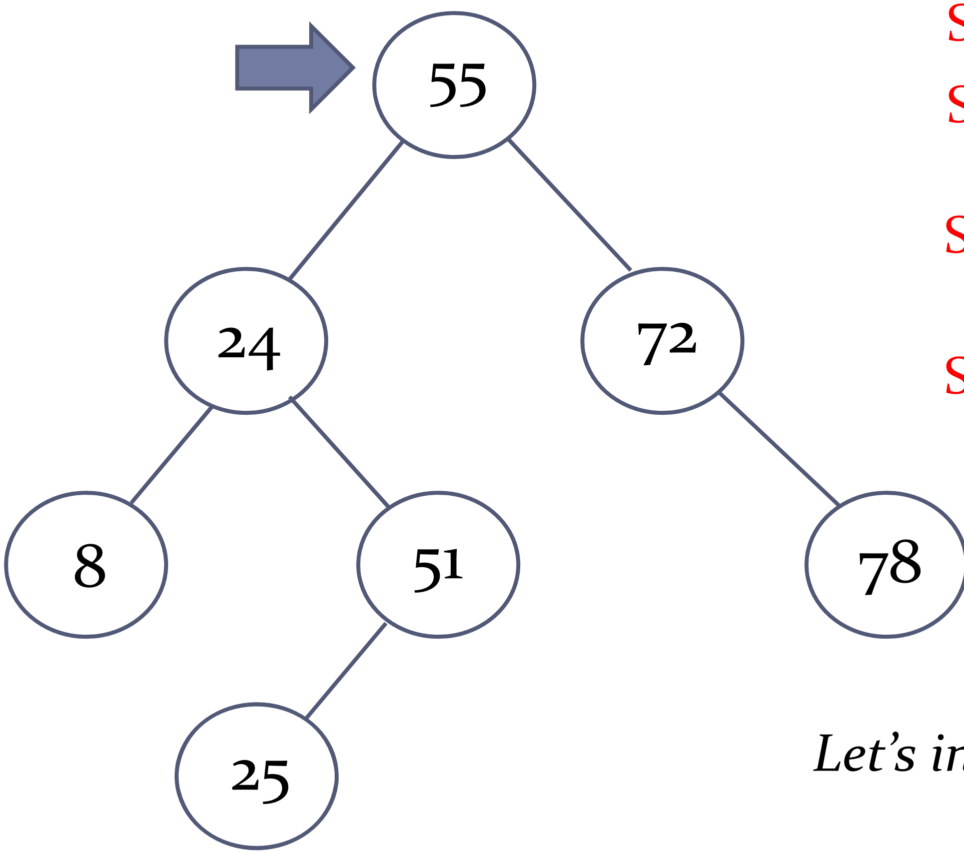

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Let's insert the following: 52

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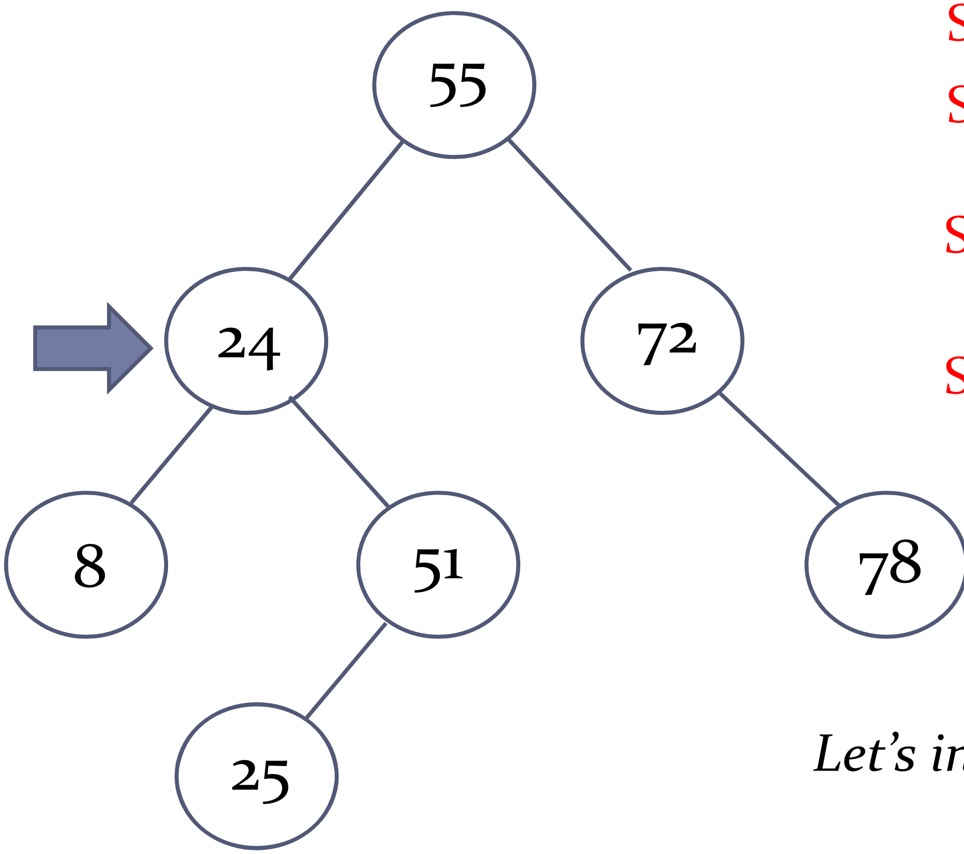

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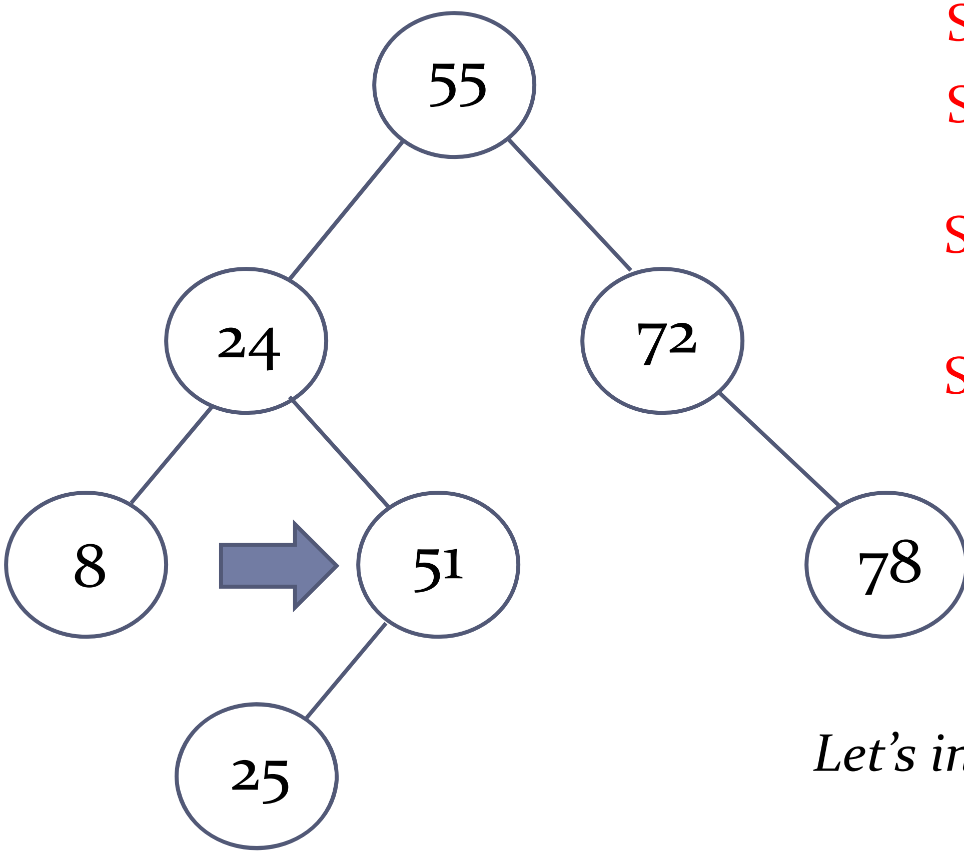

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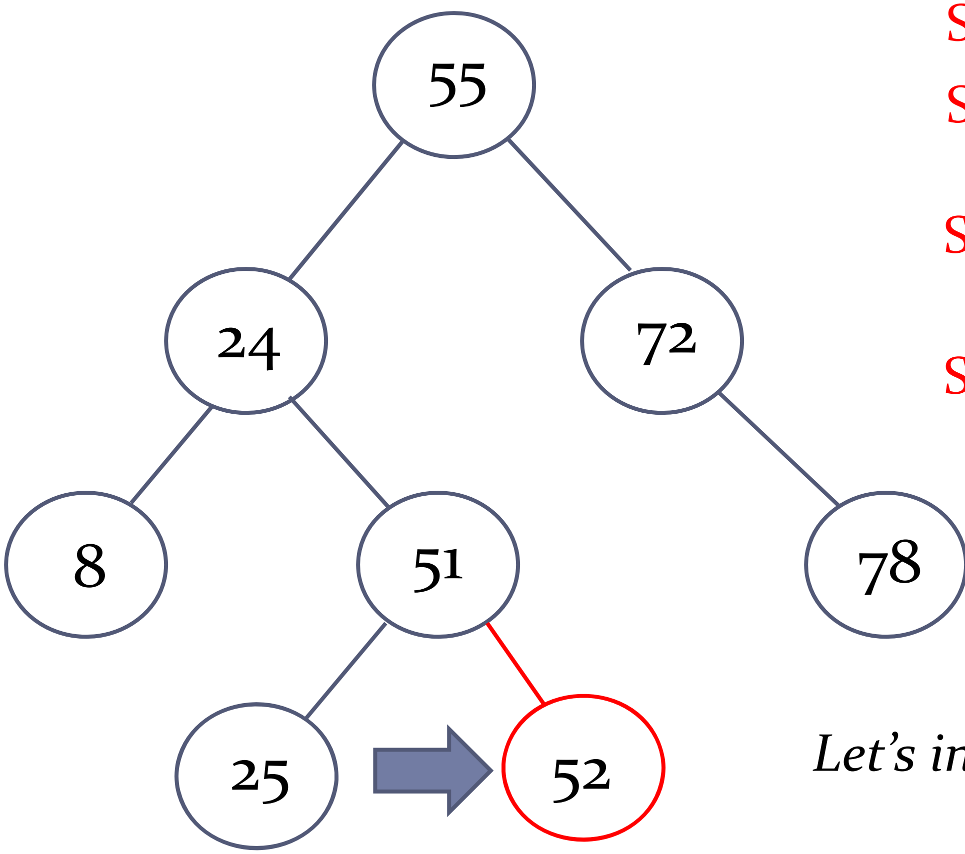

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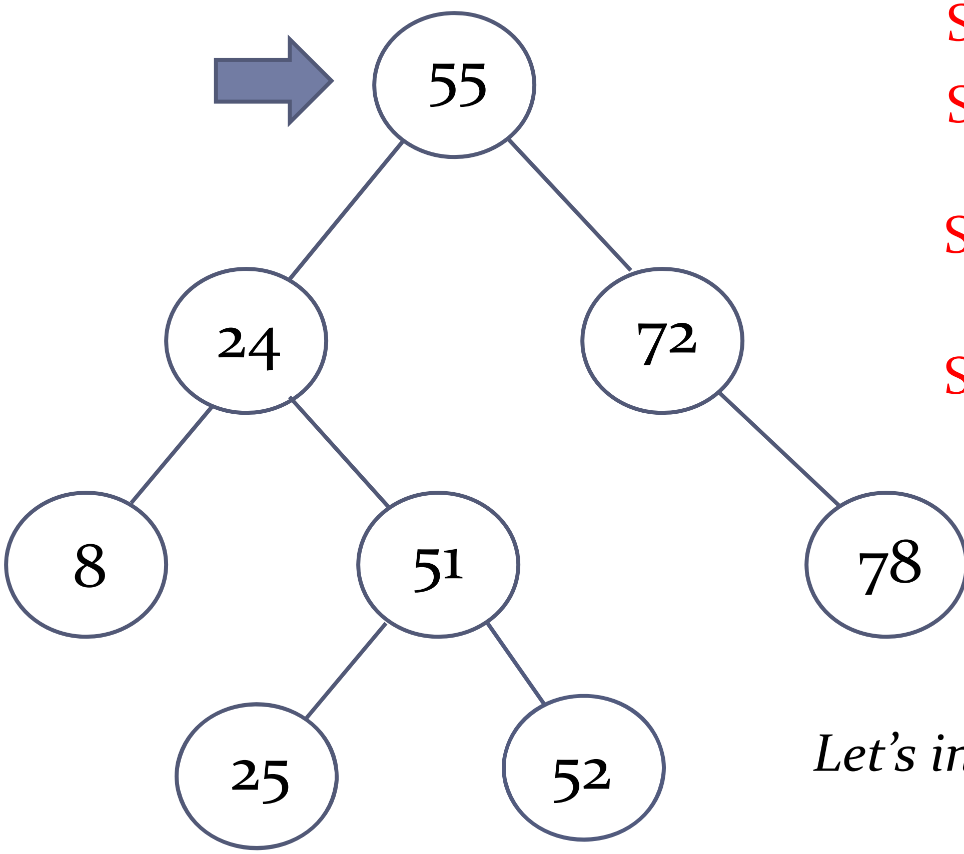

Step 1: compare node with root

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Let's insert the following: 76

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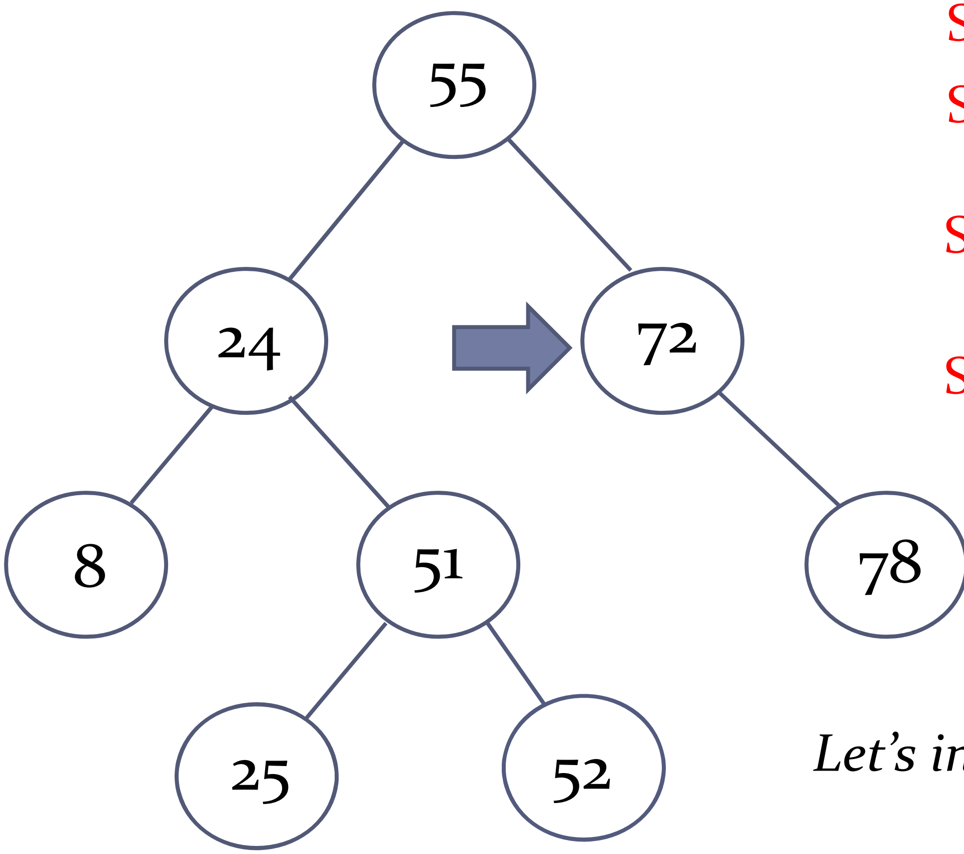

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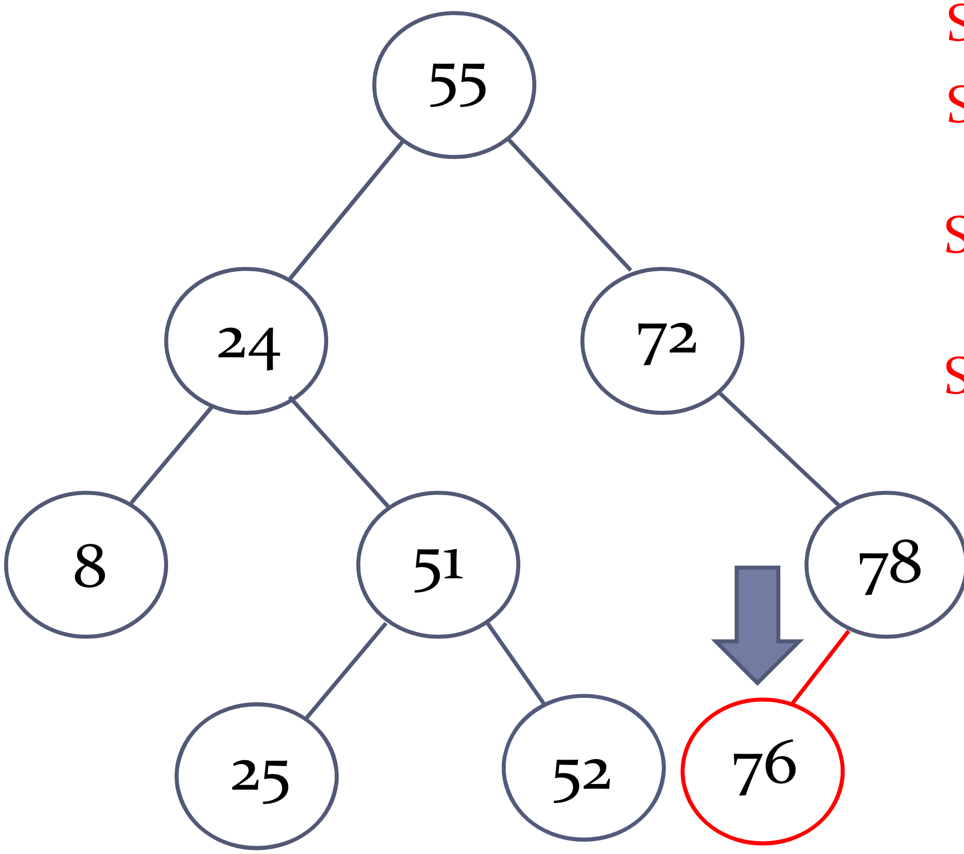

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Exercise 1 - 2

- Q1:

  Is the following a binary search tree?

```
          5
         / \
        /   \
       /     \
      /       \
     4         8
    /         / \
   /         /   \
  3         2     10
```

- Q2: What is the binary search tree created given the values in the following list:

  40, 20, 60, 10, 25, 15, 45, 70, 50